Paleomagnetism of Jurassic Rocks from Southern Alaska, and the Tectonic Implications

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Oriented samples of rocks of Jurassic age were collected from 24 separate sites located in southern and southwestern Alaska. Paleomagnetic measurements were made on these rocks, and various selection criteria were applied to obtain a mean Jurassic paleopole position of 50°N and 295°E with an α_{95} of 14.2°. This pole position is displaced from the mean paleopole of North America in the opposite direction to that predicted by Carey's orocline hypothesis for the development of Alaska. Based on this pole position, northward movement and clockwise rotation of southern Alaska must have occurred since Jurassic times, perhaps in a manner analogous to the motion of Baja California today. This Baja–Alaska model for the tectonic development of southern Alaska is currently being tested.

Des échantillons orientés de roches d'âge jurassique furent collectionnés sur 24 sites séparés situés en Alaska du Sud et du Sud-Ouest. Des mesures paléomagnétiques furent effectuées sur ces roches et des critères de selection variés furent utilisés dans le but d'obtenir une position moyenne pour le paléopôle jurassique, la position obtenue est la suivante: 50°N, 295°O avec un α_{95} de 14.2°. Cette position de pôle est déphasée par rapport au paléopole moyen de l'Amérique du nord dans une direction opposée à cette prédite par l'hypothèse de l'orocline de Carey pour ce qui est du développement de l'Alaska. En fonction de cette position de pôle, l'Alaska du Sud a du subir un mouvement vers le Nord et une rotation dans le sens horaire depuis les temps jurassiques d'une façon analogue peut-être au mouvement de Baja Califournia aujourd'hui. Ce modèle Baja–Alaska pour le développement tectonique de l'Alaska du Sud est présentement mis à l'épreuve.
[Traduit par le journal]

Introduction

The arcuate form of Alaskan topography and tectonic features has been a topic of speculation for many years, especially since the acceptance of the concepts of continental drift and plate tectonics. The most obvious of the topographic features in Alaska with an arcuate form is the Alaska Range, which is a part of the Cordillera that bounds the western edge of the Americas. The trend traced by the Coast Mountains of Canada, the Alaska Range, and the Aleutian Range curves approximately 75° in mid-Alaska. The long axis of the Brooks Range in northern Alaska forms an angle of about 30° to the trend of the Rocky Mountains. South of the Alaska Range the Saint Elias Mountains, the Chugach Mountains, and the Kenai Mountains also form an arcuate belt along the Gulf of Alaska. There are less obvious topographic features that parallel the trends of the Alaska Range and the Brooks Range, but few other features are continuous across Alaska. In terms of tectonic features, the right-lateral strike-slip Denali fault system is well expressed topographically, and coincides with the curvature of the Alaska Range. This fault system consists of individual relatively straight fault segments that extend from Chatham Strait to Bristol Bay (Grantz 1966). The Tintina and the Kaltag faults may also form an arcuate fault system, located approximately 150 miles north of the Denali fault system (Churkin 1970).

Carey (1958) explained the arcuate tectonic form of Alaska by the orocline concept, the various features being formed on the compressional side of a rift he called the Arctic sphenochasm. This rift has an opening of about 30° distributed into two *en echelon* tension features. Carey's pivot point for the Alaskan orocline is approximately 148°W and 64°N, which is just north of the bend in the Alaska Range.

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Grantz (1966) in his study of strike-slip faults in Alaska restores the movement along the faults on the western side of the bend with a simple mechanical model. He demonstrates how the bend can be straightened by rotations of up to 50°. Fifty degrees is also approximately the amount by which the Mesozoic geosynclines and geanticlines described by Payne (1955) are bent. Grantz (1966) also suggests that the arcuate tectonic form of Alaska might be one limb of a buckle created by longitudinal drift of North America toward Siberia and not by a simple rotation about a pivot in Alaska, as was Carey's (1958) hypothesis.

Geologic evidence cited by Churkin (1970) suggests that at least northern Alaska and Chukotka have been connected since Paleozoic time and probably since the Precambrian. This, of course, precludes any large-scale continental drift between northern Alaska and Siberia. Churkin (1969) also proposed that the bend in Alaska is largely the result of a long period of regional stress created by relative motion of the Pacific sea floor towards the continental margin of the Gulf of Alaska.

The arcuate Alaskan form thus has several possible causes, each involving different amounts and directions of relative motion. As a first step in an attempt to resolve the problem, paleomagnetic measurements were made on a number of Jurassic rocks from Alaska. As discussed below, the data from these measurements points to yet another model for the tectonic development of Alaska.

Techniques

Oriented cores were collected in groups, referred to here as sites, with an attempt to collect three cores or more from each stratigraphic level, and to sample as many levels as time and conditions permitted. Geographic spacing was achieved by collecting at more than one site within the same rock unit whenever possible. In general, ten cores were collected per site, with never less than six. If individual groups of cores were one mile or less from each other they are considered together as a site. When the groups of cores within the same rock unit from the same sampling area have a separation of greater than one mile then the means were considered as individual sites, unless noted otherwise. Cores were taken at least one-quarter mile from known faulting in an attempt to avoid local rotations, although in many areas in Alaska, geologic mapping or exposure is insufficient to assess this properly.

The samples were measured on a 5 Hz Schonstedt spinner magnetometer (model SSM-1A). The samples were magnetically 'cleaned' by tumbling about two mutually perpendicular axes in field free space, while being demagnetized in successively higher 60 Hz alternating magnetic fields. The 'best' AF demagnetization level was taken to be that where the scatter of individual data points was a minimum after the mean direction stopped changing significantly. A subjective estimate of the stability of the rocks was also made on the basis of whether or not the mean NRM (Natural Remanent Magnetization) for a group of samples was aligned parallel to the present or axial field before corrections for local and regional dip were made (NRM P). Low field bulk susceptibility was also measured using a susceptibility bridge technique (Collinson *et al.* 1963). All results are plotted on Wulff stereographic projections.

Results

Alaska Peninsula

Geologic Setting of the Jurassic of the Alaska Peninsula

The best geologic summary of the Alaska Peninsula available to date is that of Burk (1965). Basically, volcanic-rich rock units of early Jurassic age conformably overlie late Triassic rocks. Along the southern coast of the Alaska Peninsula the fossiliferous early Jurassic sediments consist of tuffaceous sandstone, calcareous sandstone and shale, limestone, and volcanic conglomerate grading upward to fine-grained shale and sandstone. Inland, early Jurassic porphyritic flows, breccia, and tuff make up much of the section.

A very complete section of middle and late Jurassic sedimentary rocks is well-exposed in the Tuxedni Bay region, and conformably overlies the early Jurassic. Part of this section, the Tuxedni group, is composed of siltstone, arkose, graywacke, sandstone, conglomerate, massive sandstone, and shale (Detterman and Hartsock 1966). Conformably overlying the

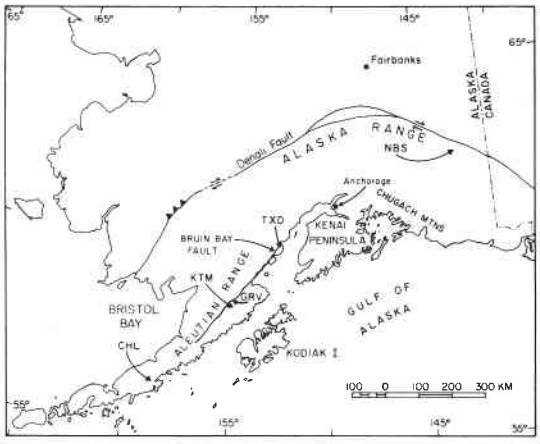


FIG. 1. Location map.

Tuxedni group is the Chinitna Formation, a massive to thinly bedded siltstone and sandstone. The upper Jurassic Naknek Formation conformably overlies the Chinitna Formation, and is composed of a basal conglomerate, arkosic sandstone, and siltstone. The Naknek Formation forms a distinctive marine rock type and is well exposed over the entire length of the Alaska Peninsula, whereas the early Jurassic formations only occur locally. Southwest of Wide Bay the late Jurassic–early Cretaceous Staniukovich Formation conformably overlies the Naknek. The Staniukovich Formation is composed of fine-grained feldspathic sandstone and arkose.

The middle Jurassic through the early Cretaceous sedimentary section, on the southeastern shore of the Alaska Peninsula, represents sediments deposited in a rapidly subsiding area close to an eroding granitic source. The Aleutian Range batholiths of Jurassic age (Reed and Lanphere 1969), though separated from the sedimentary sequence by the Bruin Bay fault, appear to have served as the source area. The Bruin Bay fault, a high-angle reverse fault upthrown on the northwest side, uplifted the granitic body relative to the Jurassic sediments. On the basis of the sedimentary record, the fault must have become active in mid- to late-Jurassic time. The Jurassic batholiths, southwest of Wide Bay, are covered by Tertiary and Quaternary age rocks.

Uplift and minor deformation must have occurred on the Alaska Peninsula in mid-Cretaceous time, because all late Cretaceous rocks rest unconformably on early Cretaceous rocks. The Tertiary is characterized by an almost complete record consisting dominantly of volcanics and volcanogenic sediments. Slight deformation and uplift in early Paleocene and in late Oligocene–early Miocene correspond in time with regional intrusive activity.

Deformation consisting of anticlinal and synclinal folding, and local faulting associated with the anticlinal complexes, culminated in the Pliocene. Northwest of the Bruin Bay fault, folding is local and generally restricted to early Mesozoic rocks, and is thought to be due to the intrusion of the Jurassic Aleutian Range batholiths. Volcanism has continued from the Pliocene to the present, and is continuing along the length of the Alaska Peninsula. Quaternary and recent flows, associated pyroclastic material, glacial material, and stream deposits make up the recent rock record on the Alaska Peninsula.

Sites on the Alaska Peninsula

Chignik Lake (CHL)—Fifty-two cores from the Upper Jurassic Naknek Formation were collected at five sites along the northeast shore of Chignik Lake (CHL-1 to 5). These sites are separated by approximately four miles. It was not possible to estimate the total number of stratigraphic feet represented by all five sites; however, each site represents at least 10 stratigraphic feet. The samples collected consist of fine- to coarse-grained sandstone. The beds for the three sites at the northwest end of the lake have a 5° to 15° dip to the northeast, and the two sites at the southeast end of the lake have dips of 10° to the southwest and southeast, respectively.

The mean vector directions and pole positions are given in Table 1 and Figs. 2 and 3. The mean NRM direction with respect to the present horizontal (NRM P) does not lie along the present axial dipole field direction for any of the sites.

Katmai area (KTM)—Cores of the Jurassic Naknek Formation and a Jurassic quartz diorite batholith were collected in the Katmai area. Twenty-six cores of gray fine-grained sandstone were collected from the Naknek Formation exposed on the lower part of the south flank of Mt. Katolinat. At this location the Naknek is intruded by rhyolite dikes and a hornblende andesite sill; the sill crosscuts the

TABLE 1. Paleomagnetic directions and pole positions

LOCATION	TREATMENT	N	DIRECTION D	I	A95	K	POLE POSITION LAT	ELONG	A95	K
CHL-1 - CHIGNIK LAKE, ALASKA (56.3N,201.2E)	NRM P	10	26	60	10.6	21.5	67	322		
	NRM	10	18	69	10.7	21.5	78	285	16.5	9.5
1 SITE UPPER JURASSIC NAKNEK FM	AF 95 OE	10	1	70	6.8	51.2	89	339	10.8	21.0
CHL-2 (56.3N,201.1E)	NRM P	10	54	60	14.2	12.6	52	292		
	NRM	10	54	55	14.2	12.6	50	296	18.4	7.8
1 SITE	AF 95 OE	10	62	61	13.5	13.8	49	282	19.5	7.1
CHL-3 (56.3N,201.1E)	NRM P	11	32	61	16.4	8.7	65	313		
	NRM	11	49	59	16.4	8.7	56	291	22.6	5.0
1 SITE	AF 95 OE	11	47	58	29.4	3.4	57	297	33.3	2.8
CHL-4 (56.3N,201.1E)	NRM P	9	77	53	15.5	12.0	34	281		
	NRM	9	81	38	15.5	12.0	24	265	14.7	13.2
1 SITE	AF 95 OE	9	90	32	23.1	5.9	15	280	19.3	8.1
CHL-5 (56.3N,201.1E)	NRM P	12	63	59	15.0	9.3	47	286		
	NRM	12	70	45	15.0	9.4	33	290	17.7	7.0
1 SITE	AF 95 OE	12	81	39	16.6	7.8	23	284	18.0	6.8
	AF 190 OE	12	70	34	19.3	6.9	29	294	18.2	6.7

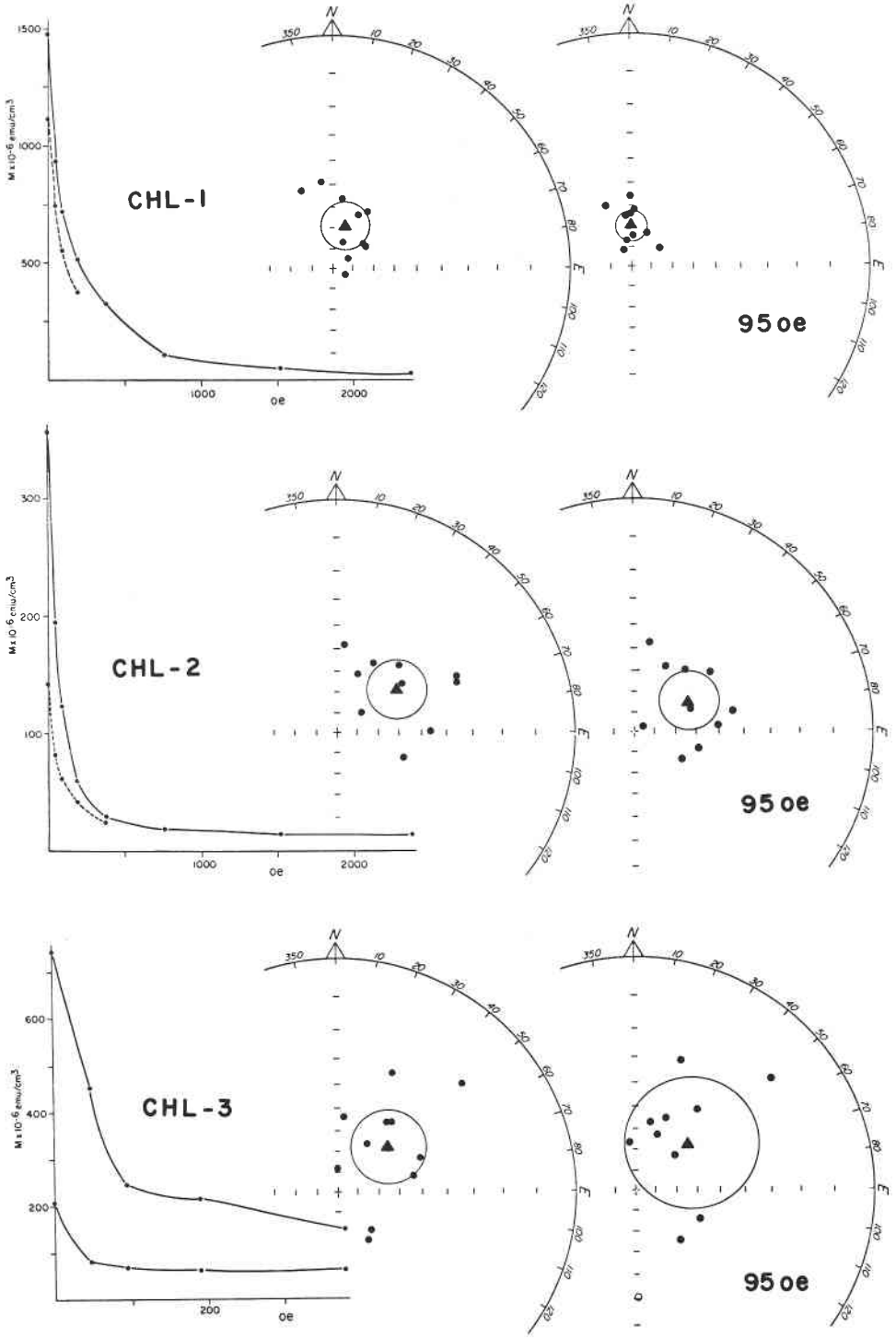
CHL-1 TO 5 MEAN NRM INTENSITY 560X10-6 EMU/CM3, SUSCEPTIBILITY 3000X10-6 EMU/CM3

NOTES:
LOCATION: local name and approximate latitude and longitude of collection locality, number of individual sites, and age and name of unit from which the samples were collected.
TREATMENT: NRM P—mean NRM direction with respect to present horizontal (not given if bedding is horizontal).
NRM—mean NRM vector direction with respect to ancient horizontal.
AF—alternating field demagnetization levels in oersteds (peak to peak).
N: number of samples used in the determination of the mean directions.
DIRECTION (of the N end of the magnetization vector):
D—mean declination in degrees east of true north.
I—mean inclination in degrees below (+) or above (−) horizontal.
A95, K: radius in degrees of the 95% circle of confidence about the mean paleomagnetic field direction; and K, the Fisher parameter, k.
POLE POSITION:
LAT, ELONG—coordinates of the mean virtual geomagnetic north pole position in degrees north (+) or south (−) latitude, and degrees east longitude.
A95, K—radius in degrees of the 95% circle of confidence about the mean paleomagnetic pole position; and K, the Fisher parameter, k.

dikes which were also sampled. The exact age of these intrusives is unknown, as the rocks are unsuitable for dating by the K–Ar method. The cores were collected at three different sites which have a separation of approximately one-half mile and represent approximately 700 stratigraphic feet (213 m) of section. The Naknek Formation dips at 3° to the east-northeast throughout the section. Six cores were also collected on cliffs above Margot Creek to the south-southeast of Mt. Katolinat. These samples represent approximately 10 ft (3 m) of stratigraphic section in flat-lying beds of fine-grained sandstone.

Mean NRM P measurements for the entire Katmai area include the present axial dipole field direction within their 95% confidence limit. The mean labelled KTM-1 (Fig. 4, Table 2) represents samples of the Naknek Formation only. Cores taken from these sediments were at least six feet (1.8 m) from the baked zone surrounding the intrusives and do not appear to have been altered magnetically by them. Demagnetization in an AF of 95 Oe removed a component along the direction of the present field, and minimized the scatter.

Demagnetization of the remaining samples (KTM-2, table 2) at an AF of 48 Oe removed a present field component and reduced the scatter. Subsequent demagnetization up to 380 Oe did not significantly change the position of the mean; however, further demagnetization displaced the mean significantly and increased the scatter about it. Because the mean magnetic direction after demagnetization at 1500 Oe was apparently still moving, the stability of the samples is in question and the results have not been included in the overall means.


CHL-1
95 oe
CHL-2
95 oe
CHL-3
95 oe
N
E
0e

Also from the Katmai area, twenty-seven samples of mid Jurassic age (Reed and Lanphere 1969) quartz diorite were collected from the Aleutian Range batholith along the shores of the Iliuk Arm of Naknek Lake. Two sites were on the north shore, at the base of Mt. La Gorce, and about 3.5 miles (5.6 km) apart. A third site was located on an island on the south side of the lake. These sites are a mile (1.6 km) or more from the trace of the Bruin Bay fault which is upthrown on the northwest (Keller and Reiser 1959). No evidence of attitude was observed in this reconnaissance study of the area, and it has been assumed that the batholith has not tilted or rotated significantly since its emplacement. Demagnetization of these samples (KTM-3, Table 2) in an AF of 190 Oe failed to shift the mean significantly away from the present field direction. In an AF of 380 Oe the samples became unstable and exhibited a lack of repeatability of measurements.

Lake Grosvenor (GRV)—Ten cores of early Jurassic (Reed and Lanphere 1969) granodiorite were collected from the southern shore of Lake Grosvenor (GRV-1) about 6 miles (9.7 km) east of its connection to Lake Colville. Samples were taken over a distance of approximately 100 ft (30.5 km). Thirteen cores of the same early Jurassic granodiorite (GRV-2) were collected about 3 miles (4.8 km) east of site GRV-1. Samples were collected over approximately 150 ft (45.7 m). The batholith is again assumed to have the same attitude now as during magnetization.

At GRV-1 the mean NRM P direction does not coincide with the present axial dipole field direction. After demagnetization a significant shift in the mean vector direction and marked reduction in scatter occurred (Fig. 5, Table 3). At GRV-2 the mean NRM P vector was in the same direction as that due to the present axial dipole field. After demagnetization the mean direction shifted away from the present axial field, probably indicating the removal of a secondary component along the present field direction (Fig. 5, Table 3).

Nine cores of the upper Jurassic Naknek Formation, Chisik Conglomerate member, were collected from the southern shore of Lake Grosvenor (GRV-3), approximately 3.5 miles (5.6 km) east of GRV-2 and approximately ¾ of a mile (1.2 km) southeast of the Bruin Bay fault. About 30 ft (9.1 m) of medium-grained arkosic sandstone, included locally in the conglomerate, were sampled. The beds are massive, and dip west-southwest at 5°. These samples had a totally unstable magnetic signature. The NRM and AF levels of 95 and 190 Oe had nearly a random distribution of vector directions. These results have been discarded for the purpose of this investigation.

Seven cores of the Chisik Conglomerate member of the upper Jurassic Naknek Formation were also collected from the southern shore of Lake Grosvenor (GRV-4) about 1.5 miles (2.4 km) south of GRV-3. These samples come from a siltstone bed included in the conglomerate. It is approximately 20 ft (6.1 m) thick, is cross-bedded, and abundantly fossiliferous. These beds dip west-northwest at 4°. The mean NRM P direction for this site does not lie along the present axial dipole field direction. After demagnetization the scatter of

FIG. 2. Paleomagnetic data from Chignik Lake, Alaska. From left to right: Magnetic intensity during AF demagnetization; NRM vector directions; and vector directions after AF demagnetization.

Paleomagnetic data for particular sampling localities are shown according to the following notation:

$M \times 10^{-6}$emu/cm^3—Magnetic intensity vs. alternating field demagnetization level (oersteds, peak to peak).

NRM—NRM vector directions of the samples and their mean, with respect to ancient horizontal.

Demagnetization—Vector directions of the samples and their mean after AF demagnetization at the indicated level (oersteds, peak to peak).

North ends of vectors are shown on Wulff stereographic projections with solid circles indicating lower (+) hemisphere and open circles indicating upper (−) hemisphere. A triangle, solid or open as above, indicates the mean vector directions; a circle of radius α_{95} is drawn about the mean. Mean data are from relevant table.

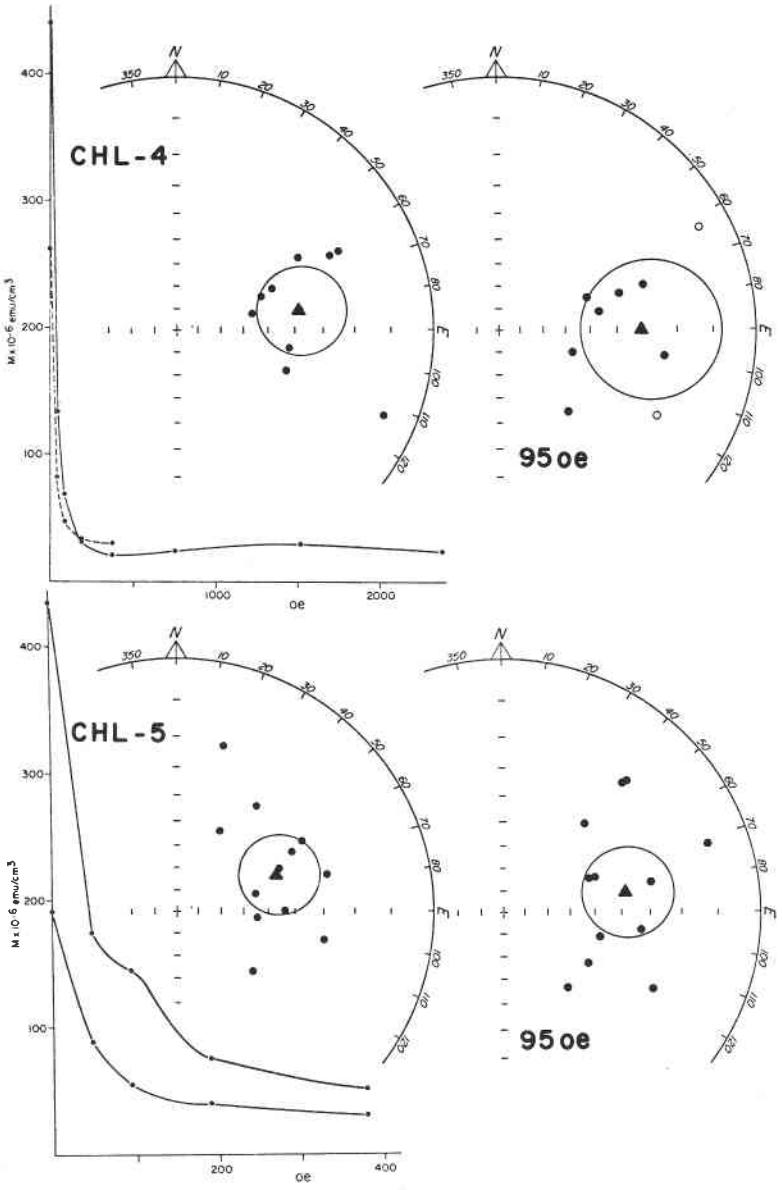


FIG. 3. Paleomagnetic data from Chignik Lake, Alaska. From left to right: Magnetic intensity during AF demagnetization; NRM vector directions; and vector directions after AF demagnetization. See Fig. 2 for explanation of notation.

data points was slightly reduced, possibly indicating the removal of secondary components (Fig. 5, Table 3).

Tuxedni Bay (TXD)—At 10 sites, spread over approximately 23 miles (37 km), over 140 cores from the Naknek Formation, Tuxedni Group, Chinitna Formation, and a quartz-diorite batholith, all of Jurassic age, were collected along the shores of Tuxedni Bay, and from the Hickerson Lake, Tuxedni Glacier, and Tuxedni River regions.

The first site (TXD-1), located across Tuxedni Channel from Chisik Island, covers 80 stratigraphic feet (24.4 m) of section and ranges from fine- to coarse-grained arkosic sandstone. The 15 samples from TXD-1 are

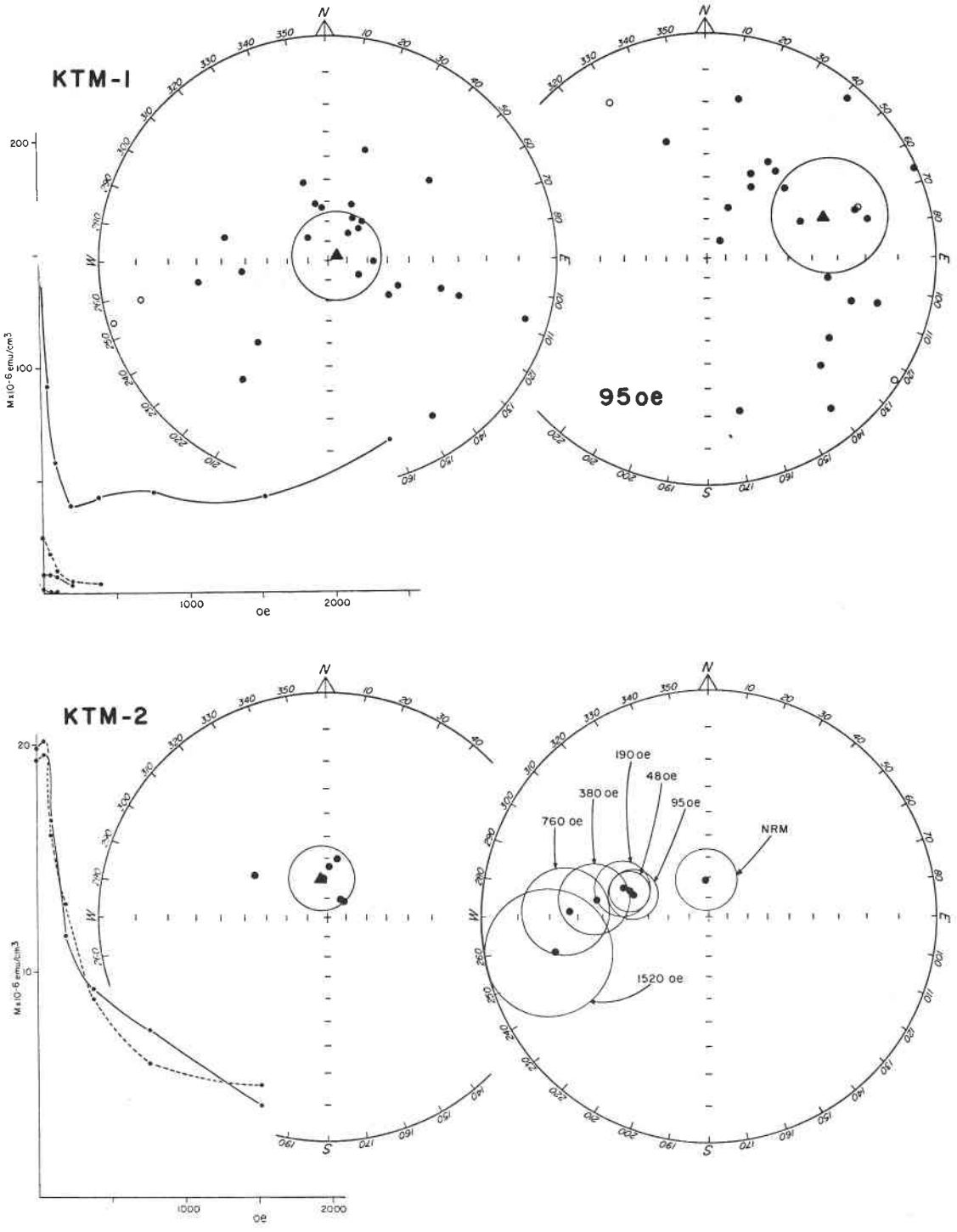


FIG. 4. Paleomagnetic data from Katmai, Alaska. From left to right: Magnetic intensity during AF demagnetization; NRM vector directions; and vector directions after AF demagnetization. See Fig. 2 for explanation of notation.

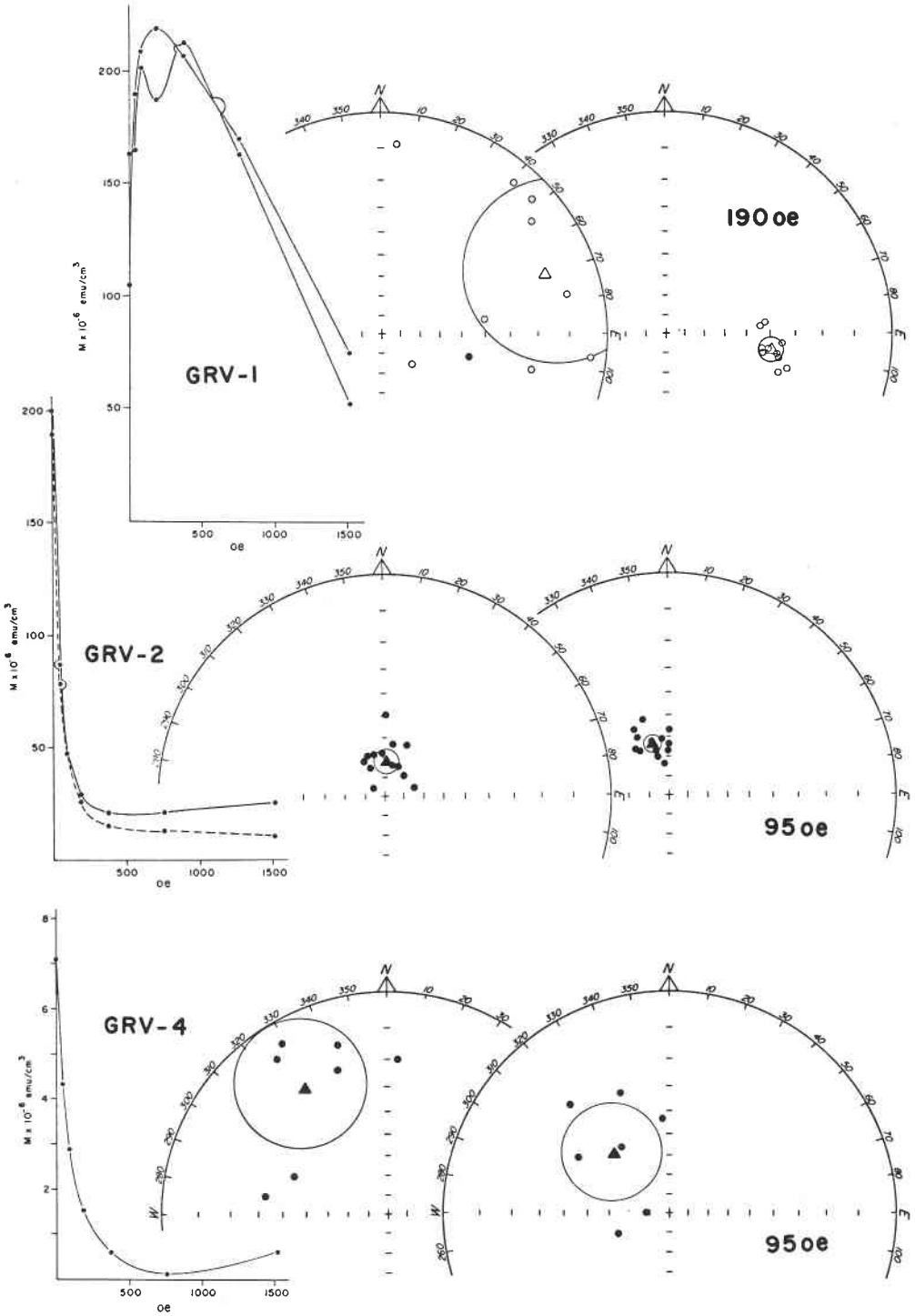


FIG. 5. Paleomagnetic data from Grosvenor Lake, Alaska. From left to right: Magnetic intensity during AF demagnetization; NRM vector directions; and vector directions after AF demagnetization. See Fig. 2 for explanation of notation.

TABLE 2. Paleomagnetic directions and pole positions

LOCATION	TREATMENT	N	DIRECTION D	I	A95	K	POLE POSITION LAT	ELONG	A95	K
KTM-1 - KATMAI AREA, ALASKA		26	69	87	21.3	2.8	60	215		
(58.4N,204.6E)	NRM	26	64	85	21.3	2.8	66	231	29.6	1.9
3 SITES	AF 48 OE	24	38	12	46.8	1.4	28	332	41.1	1.5
UPPER JURASSIC NAKNEK FM	AF 95 OE	24	72	34	21.8	2.8	29	301	21.8	2.8
	MEAN NRM INTENSITY		30X10-6 EMU/CM3, SUSCEPTIBILITY				90X10-6 EMU/CM3			
KTM-2	NRM	6	354	72	15.3	23.1	89	86	23.5	9.1
(58.4N,204.6E)	AF 48 OE	6	288	50	9.2	54.3	37	118	9.9	47.1
1 SITE	AF 95 OE	6	286	52	10.6	41.0	37	120	12.1	31.5
UPPER JURASSIC NAKNEK FM	AF 190 OE	6	288	47	12.0	31.9	34	115	11.3	36.7
	AF 380 OE	6	278	37	13.8	24.4	22	119	12.2	30.9
	AF 760 OE	6	272	27	15.9	18.7	14	120	13.4	26.1
	AF 1520 OE	6	257	21	20.5	11.6	4	131	17.2	16.1
	MEAN NRM INTENSITY		20X10-6 EMU/CM3, SUSCEPTIBILITY				275X10-6 EMU/CM3			
KTM-3	NRM	27	349	67	11.2	7.1	81	82	15.5	4.2
(58.5N,204.5E)	AF 190 OE	27	333	65	14.1	4.9	72	109	18.4	3.3
3 SITES										
MID JURASSIC GRANODIORITE	MEAN NRM INTENSITY		310X10-6 EMU/CM3, SUSCEPTIBILITY				3000X10-6 EMU/CM3			

NOTE: For explanation, see notes to Table 1.

TABLE 3. Paleomagnetic directions and pole positions

LOCATION	TREATMENT	N	DIRECTION D	I	A95	K	POLE POSITION LAT	ELONG	A95	K
GRV-1 - GROSVENOR LAKE, ALASKA	NRM	10	70	-15	29.5	3.7	1	314	26.7	4.2
(58.7N,204.7E)	AF 190 OE	10	99	-40	4.8	104.3	-24	300	5.1	92.2
1 SITE	AF 380 OE	10	103	-40	5.5	77.1	-26	297	5.5	79.0
LOWER JURASSIC DIORITE	AF 760 OE	10	101	-43	5.5	79.0	-27	299	5.4	81.6
	AF 1520 OE	10	107	-37	9.3	28.1	-27	291	9.6	26.4
	MEAN NRM INTENSITY		200X10-6 EMU/CM3, SUSCEPTIBILITY				5050X10-6 EMU/CM3			
GRV-2	NRM	13	3	72	6.4	43.5	88	322	10.3	17.1
(58.7N,204.7E)	AF 95 OE	13	344	63	4.4	93.1	73	67	6.1	46.8
1 SITE										
LOWER JURASSIC QUARTZ DIORITE										
	MEAN NRM INTENSITY		200X10-6 EMU/CM3, SUSCEPTIBILITY				4650X10-6 EMU/CM3			
GRV-3	NRM P	9	K VALUE INDICATES RANDOM SCATTER EVEN AFTER							
(58.7N,204.8E)	NRM	9	DEMAGNETIZATION AT AF 95 AND 190 OE							
1 SITE										
UPPER JURASSIC NAKNEK FORMATION										
	MEAN SUSCEPTIBILITY		30X10-6 EMU/CM3							
GRV-4	NRM P	7	327	27	23.2	7.7	40	68		
(58.6N,204.8E)	NRM	7	327	23	23.1	7.9	38	69	23.4	7.6
1 SITE	AF 95 OE	7	318	51	22.3	8.3	53	96	28.1	5.6
UPPER JURASSIC NAKNEK FM										
	MEAN NRM INTENSITY		7X10-6 EMU/CM3, SUSCEPTIBILITY				60X10-6 EMU/CM3			

NOTE: For explanation, see notes to Table 1.

from the lower sandstone member of the upper Jurassic Naknek Formation which dips east-southeast at 15°.

Samples from the Red Glacier Formation of the middle Jurassic Tuxedni Group were collected from the southwestern shore of Tuxedni Bay, approximately 1.25 miles (2.0 km) northwest of Fossil Point (TXD-2). These samples consist of calcareous fine-grained sandstone. Approximately 10 stratigraphic feet (3 m) were sampled in beds that dip at 15° east-southeast.

On the northeast shore of Tuxedni Bay less than a quarter mile from the mouth of the

Tuxedni River, 10 cores of early Jurassic, 170 to 168 my (Reed and Lanphere 1969), granodiorite were collected (TXD-3). No evidence of attitude was observed, and it has been assumed that the batholith has not tilted or rotated significantly since its emplacement.

Samples of the upper Jurassic Tonnie Siltstone member of the Chinitna Formation were collected on the westernmost tip of Chisik Island (TXD-4). These samples are dense, calcareous fine-grained siltstone containing shell fragments and occurring as lenticular bodies in a coarser grained arenaceous siltstone. The samples came from approximately 10 ft (3 m) of section in beds that dip east-northeast at 12°.

Twenty-two cores from the Pomeroy Arkose member of the upper Jurassic Naknek Formation were collected at the south end of Hickerson Lake (TXD-5). Approximately 40 stratigraphic ft (12 m) of a medium- to fine-grained arkosic sandstone were sampled. The beds dip south-southeast at 24°.

Twenty-five cores of the middle Jurassic Red Glacier Formation of the Tuxedni Group were also collected along Hungryman Creek (TXD-6). Approximately 70 stratigraphic ft (21 m) of gray to tan, fine-grained siltstone were sampled. These generally massive and uniform beds dip east-southeast at 11°. This locality is about a quarter mile (0.4 km) from a local small-displacement normal fault.

Eighteen cores of the lower sandstone member of the upper Jurassic Naknek Formation were drilled along the northeastern shore of Hickerson Lake (TXD-7). Approximately 50 stratigraphic ft (15 m) of light gray, fine-grained arkosic sandstone were collected. These beds are massive, and dip east-southeast at 32°.

Thirteen cores of the upper Jurassic Naknek Formation, Pomeroy Arkose member, were collected along the north shore of the small lake at the southwest base of Saddle Mountain (TXD-8). Approximately 60 stratigraphic ft (18 m) of medium- to fine-grained arkosic sandstone were sampled. This massively bedded member has beds that dip east-southeast at 24°.

Eight cores of the upper Jurassic Naknek Formation, Snug Harbor Siltstone member, were collected in Snug Harbor on Chisik Island (TXD-9). Approximately 25 stratigraphic ft (8 m) of fine-grained siltstone with beds locally containing small concretions were collected. These thin-bedded sediments are well fractured at most outcrops and dip east-southeast at 24°.

Nine cores of the early Jurassic, 170 to 168 my (Reed and Lanphere 1969) quartz diorite were collected from a site located above the Tuxedni Glacier (TXD-10). Petrographically these samples are the same as those collected at TXD-3 and the batholith is again assumed to have its original attitude, although with less justification, as it is located on the north side of the Bruin Bay fault with no adjacent sedimentary units. Samples were taken over approximately 100 ft (30 m).

With the exception of the samples from TXD-1, all the TXD sites have NRM P directions significantly displaced from the present field direction. On AF demagnetization the TXD-1 mean shifted away from the present field. All the sites showed some shift of mean direction and change in the amount of scatter, and the 'best' result was selected using the criteria described below. The data are presented in Table 4 and in Figs. 6, 7, and 8.

South-central Alaska

Over 25 cores from Jurassic sedimentary rocks were collected at two different sites from the Nabesna area in south-central Alaska (Fig. 1).

Geologic Setting of the Mentasta Mountains

The geologic history presented is derived largely from Richter (1967). The Mentasta Mountains are divided into two geologic sections by the Denali fault. On the north of the Denali fault, are Paleozoic, low-grade, regionally metamorphosed sediments and volcanics. On the south side, relatively unmetamorphosed volcanic and sedimentary units dip at moderate angles to the northeast. The oldest rock unit is the Permian Mankomen argillite which is unconformably overlain by Permian–Triassic amygdaloidal basalt and limestone. Diorite, granodiorite, and quartz monzonite bodies intruded the rocks throughout the Mentasta Mountain area in early Mesozoic times.

Upper Jurassic to lower Cretaceous marine sediments unconformably overlie the Triassic limestone. The Jura-Cretaceous marine sedi-

TABLE 4. Paleomagnetic directions and pole positions

LOCATION	TREATMENT	N	DIRECTION D	I	A95	K	POLE POSITION LAT	ELONG	A95	K
TXD-1 - TUXEDNI BAY, ALASKA	NRM P	15	15	67	11.1	12.9	77	341		
(60.1N,207.4E)	NRM	15	48	64	11.1	12.9	60	296	14.3	8.1
1 SITE UPPER JURASSIC NAKNEK FM	AF 95 OE	15	14	53	15.2	7.3	65	0	15.2	7.3
	MEAN NRM INTENSITY 280X10-6 EMU/CM3, SUSCEPTIBILITY 3800X10-6 EMU/CM3									
TXD-2	NRM P	14	239	38	14.4	8.6	5	154		
(60.2N,207.3E)	NRM	14	227	42	14.4	8.6	4	165	16.2	7.0
1 SITE MID JURASSIC TUXEDNI GROUP RED GLACIER FM	AF 95 OE	14	238	49	12.7	10.8	12	159	15.4	7.6
	MEAN NRM INTENSITY 45X10-6 EMU/CM3, SUSCEPTIBILITY 280X10-6 EMU/CM3									
TXD-3	NRM	10	51	52	12.1	17.0	48	314	15.2	11.1
(60.3N,207.1E)	AF 190 OE	10	36	45	18.4	7.8	50	336	20.5	6.5
1 SITE LOWER JURASSIC QUARTZ DIORITE										
	MEAN NRM INTENSITY 200X10-6 EMU/CM3, SUSCEPTIBILITY 430X10-6 EMU/CM3									
TXD-4	NRM P	9	68	57	20.6	7.2	43	294		
(60.2N,207.4E)	NRM	9	70	45	20.6	7.2	36	298	19.5	7.9
1 SITE UPPER JURASSIC CHINITNA FM TUNNIE SILTSTONE MEMBER	AF 95 OE	9	71	66	15.3	11.5	52	276	20.4	7.3
	MEAN NRM INTENSITY 4X10-6 EMU/CM3, SUSCEPTIBILITY 35X10-6 EMU/CM3									
TXD-5 - TUXEDNI BAY, ALASKA	NRM P	22	35	46	8.5	14.3	50	335		
(59.9N,207.1E)	NRM	22	63	58	8.5	14.3	48	294	12.4	7.2
1 SITE UPPER JURASSIC NAKNEK FM POMEROY MEMBER	AF 95 OE	22	85	74	6.0	27.4	51	258	10.7	9.4
	MEAN NRM INTENSITY 430X10-6 EMU/CM3, SUSCEPTIBILITY 5700X10-6 EMU/CM3									
TXD-6	NRM P	25	32	49	9.5	10.3	53	337		
(60.2N,207.3E)	NRM	25	43	45	9.5	10.3	48	327	11.0	7.9
1 SITE MID JURASSIC TUXEDNI GROUP RED GLACIER FM	AF 190 OE	25	33	61	6.9	18.8	64	323	9.8	9.7
	MEAN NRM INTENSITY 30X10-6 EMU/CM3, SUSCEPTIBILITY 350X10-6 EMU/CM3									
TXD-7	NRM P	18	31	48	9.3	14.9	52	339		
(59.9N,207.3E)	NRM	18	58	34	9.3	14.9	33	314	9.2	15.0
1 SITE UPPER JURASSIC NAKNEK FM LOWER SANDSTONE MEMBER	AF 95 OE	18	73	48	5.6	39.5	34	295	6.4	30.2
	MEAN NRM INTENSITY 430X10-6 EMU/CM3, SUSCEPTIBILITY 2700X10-6 EMU/CM3									
TXD-8	NRM P	13	47	48	15.6	8.0	46	320		
(60.0N,207.3E)	NRM	13	61	33	15.6	8.0	34	314	17.4	6.6
1 SITE UPPER JURASSIC NAKNEK FM POMEROY MEMBER	AF 95 OE	13	67	51	11.6	13.7	41	298	15.4	8.3
	MEAN NRM INTENSITY 740X10-6 EMU/CM3, SUSCEPTIBILITY 6700X10-6 EMU/CM3									
TXD-9	NRM P	8	80	50	15.8	13.3	31	289		
(60.1N,207.4E)	NRM	8	90	29	15.8	13.3	14	290	14.4	15.8
1 SITE UPPER JURASSIC NAKNEK FM SNUG HARBOR SILTSTONE	AF 95 OE	8	88	32	15.1	14.4	18	290	12.3	21.3
	MEAN NRM INTENSITY 120X10-6 EMU/CM3, SUSCEPTIBILITY 1760X10-6 EMU/CM3									
TXD-10	NRM	9	316	37	15.3	11.6	42	86	15.3	12.3
(60.1N,206.9E)	AF 190 OE	9	327	48	17.7	9.4	55	79	19.0	8.3
1 SITE LOWER JURASSIC QUARTZ DIORITE	AF 380 OE	9	264	77	34.2	3.2	53	167	46.7	2.2
	AF 760 OE	9	328	48	21.4	6.7	56	81	22.8	6.0
	AF 1520 OE	9	MEASUREMENTS UNREPEATABLE							
	MEAN NRM INTENSITY 220X10-6 EMU/CM3, SUSCEPTIBILITY 4600X10-6 EMU/CM3									

NOTE: For explanation, see notes to Table 1.

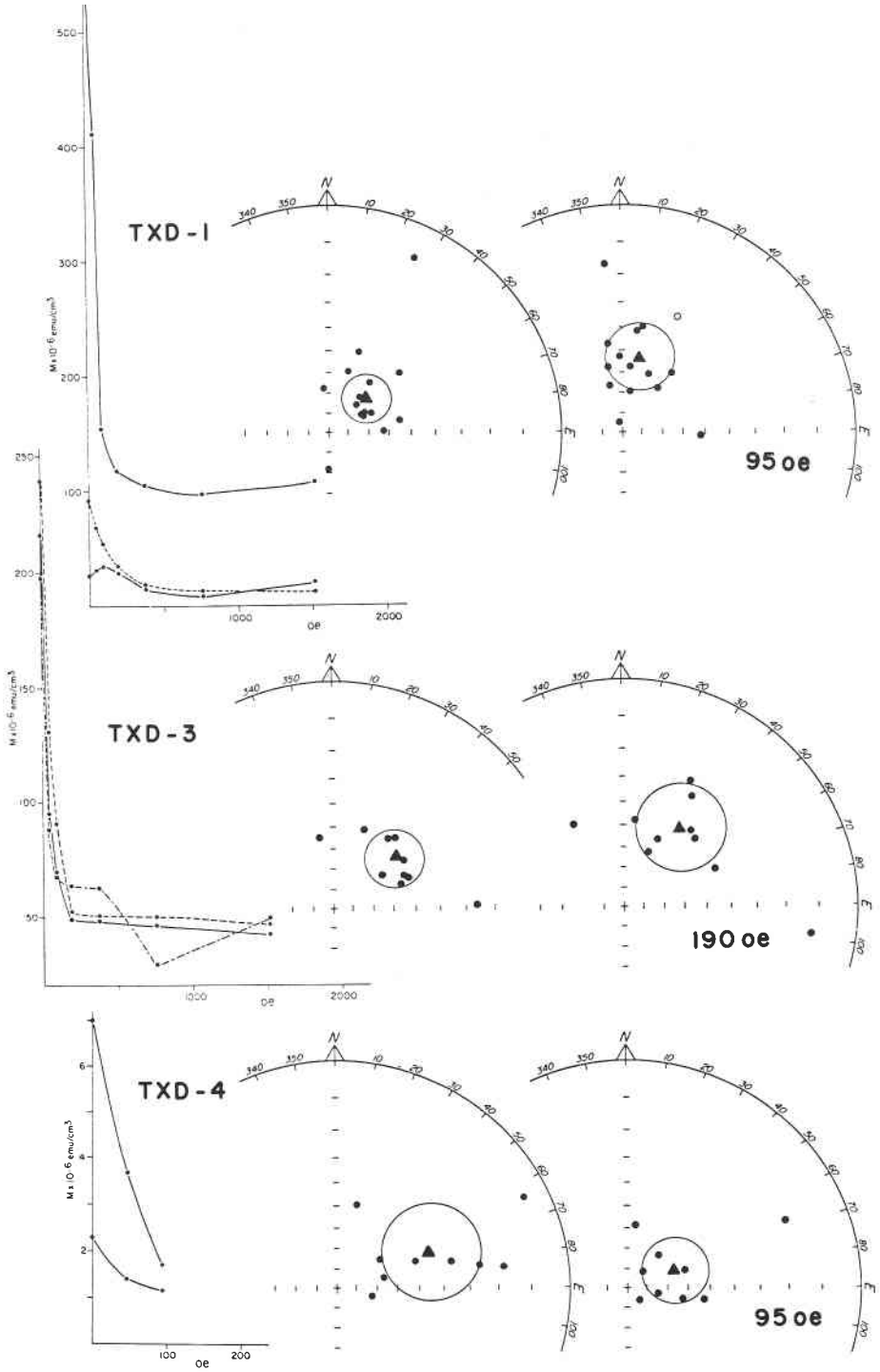


FIG. 6. Paleomagnetic data from Tuxedni Bay, Alaska. From left to right: Magnetic intensity during AF demagnetization; NRM vector directions; and vector directions after AF demagnetization. See Fig. 2 for explanation of notation.

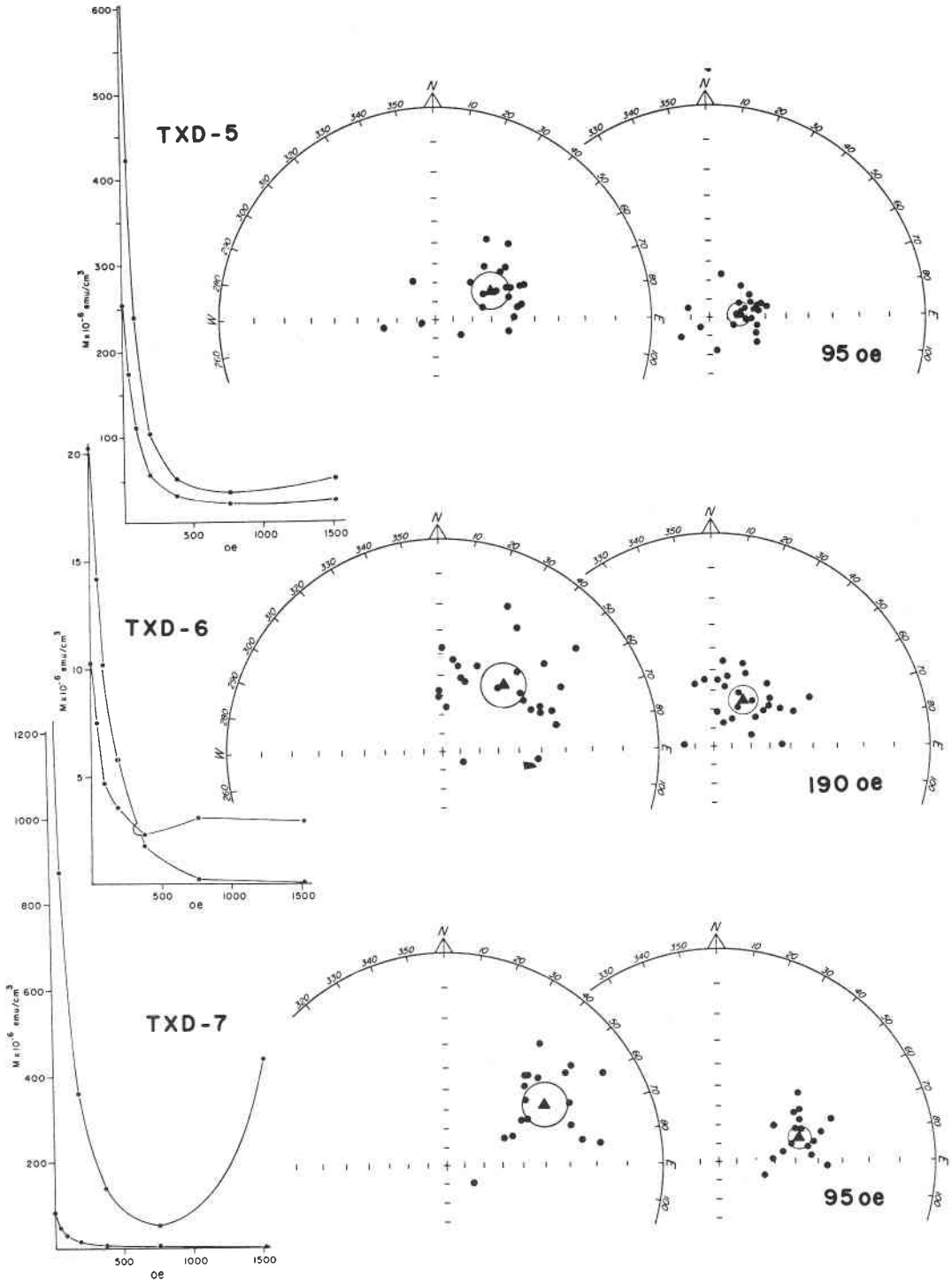


FIG. 7. Paleomagnetic data from Tuxedni Bay, Alaska. From left to right: Magnetic intensity during AF demagnetization; NRM vector directions; and vector directions after AF demagnetization. See Fig. 2 for explanation of notation.

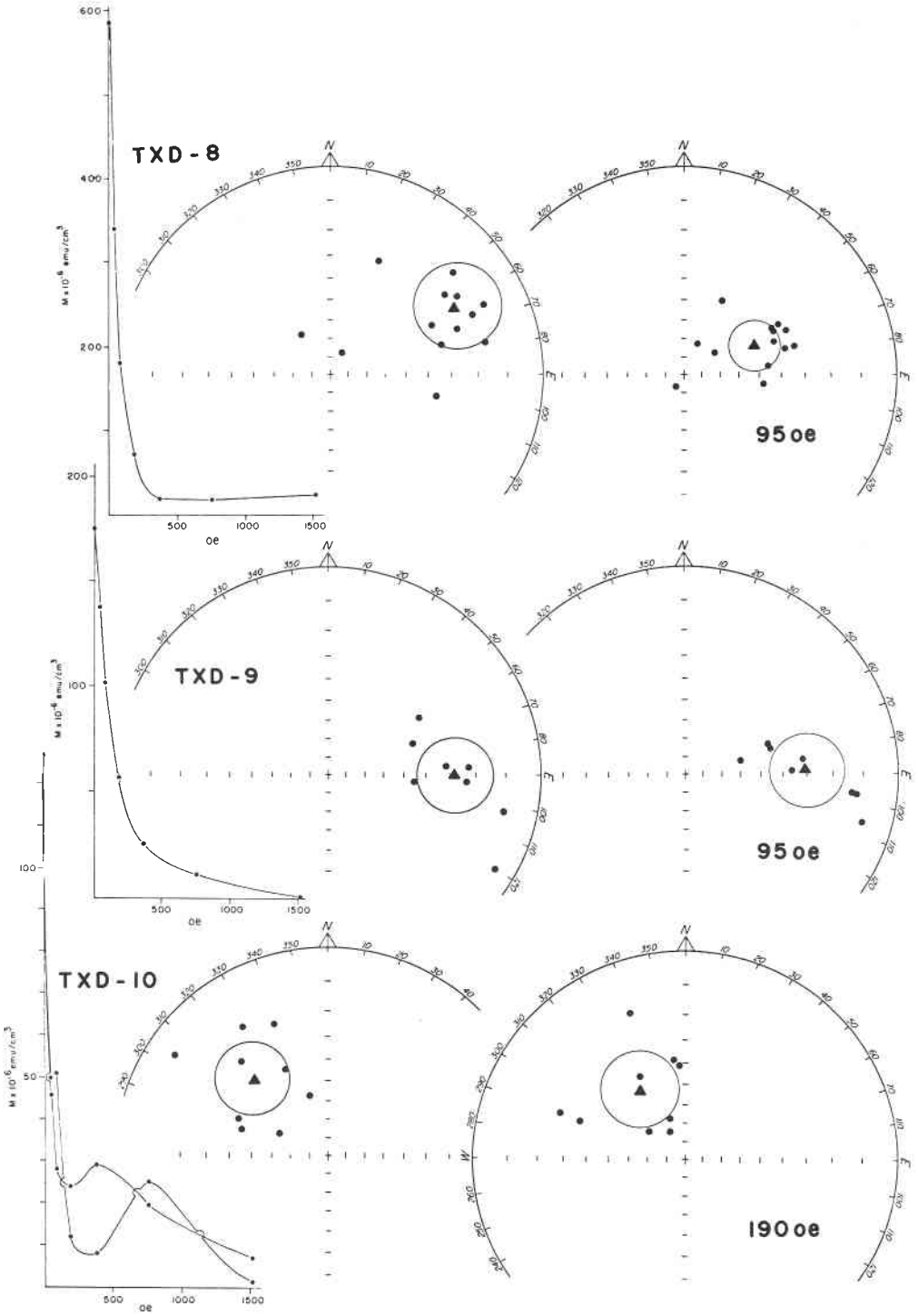


FIG. 8. Paleomagnetic data from Tuxedni Bay, Alaska. From left to right: Magnetic intensity during AF demagnetization; NRM vector directions; and vector directions after AF demagnetization. See Fig. 2 for explanation of notation.

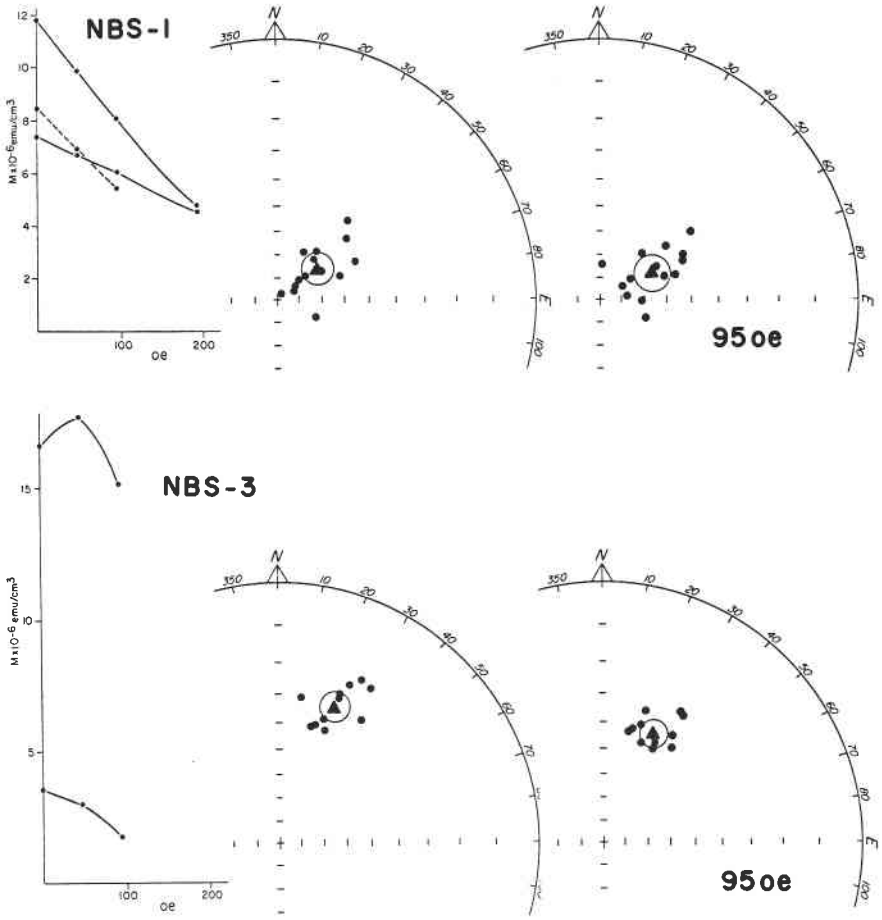


FIG. 9. Paleomagnetic data from Nabesna, Alaska. From left to right: Magnetic intensity during AF demagnetization; NRM vector directions; and vector directions after AF demagnetization. See Fig. 2 for explanation of notation.

ments are part of a unit that extends into the Nutzotin Mountains to the southeast and into areas of the Alaska Range to the northwest. The Denali fault trends northwest and has probably been active since the middle Mesozoic with most of the offset occurring during early Tertiary time. Right-lateral strike-slip has been the principal direction of motion, but evidence for a vertical displacement is also observed. South of the Denali fault, major folding is only evident close to the fault zone.

Sites in the Mentasta Mountains

Nabesna area (NBS)—Jura-Cretaceous marine sediments were collected from two sites on either side of Platinum Creek. The samples are fine-grained calcareous sandstones. The sandstone occurs interbedded in graywacke. NBS-1 on the east side of Platinum Creek covers 400 ft (122 m) of section in beds dipping 10° to 24° east-northeast. On the west side, NBS-3, approximately 50 ft (15 m) of section is represented in beds that dip northeast at 17°.

The means of the measured paleomagnetic field directions and paleomagnetic pole positions are shown in Fig. 9 and Table 5. NBS-1, from Jurassic marine sediments, has a mean NRM P which includes the present dipole field direction within its 95% confidence limits. After demagnetization of an AF of 48 Oe, a component in the present field direction was removed. The mean NRM P for the sediments from NBS-3 does not lie along the present field direction. The samples from both NBS-1 and 3

TABLE 5. Paleomagnetic directions and pole positions

	LOCATION	TREATMENT	N	DIRECTION D	I	A95	K	POLE POSITION LAT	ELONG	A95	K
NBS-1	NABESNA AREA, ALASKA (62.6N,217.0E)	NRM P	14	10	75	5.2	60.4	85	312		
		NRM	14	55	69	6.9	34.2	61	297	10.2	16.2
	1 SITE JURASSIC MARINE SEDS	AF 95 OE	14	63	65	7.4	30.2	54	297	10.6	15.1
		MEAN NRM INTENSITY		10X10-6 EMU/CM3, SUSCEPTIBILITY				35X10-6 EMU/CM3			
NBS-3	(62.6N,217.0E)	NRM P	11	2	61	5.5	69.3	70	33		
		NRM	11	23	31	5.5	69.2	42	7	4.7	94.2
	1 SITE JURASSIC MARINE SEDS	AF 95 OE	11	26	40	4.7	93.3	47	1	4.8	91.4
		MEAN NRM INTENSITY		10X10-6 EMU/CM3, SUSCEPTIBILITY				30X10-6 EMU/CM3			

NOTE: For explanation, see notes to Table 1.

had intensities too small to demagnetize beyond the 95 Oe level. The scatter for both sites was extremely small.

Summary of Results

The mean virtual geomagnetic poles (VGP), corresponding to the natural remanent magnetization with respect to ancient horizontal (NRM), and with respect to present horizontal (NRM P), and also after various AF demagnetization levels, are plotted in Fig. 10. The only sites excepted are KTM-3 and GRV-3, which were totally unstable. Selection criteria have been applied to the results shown in Fig. 10 by discarding sites CHL-3, KTM-1, and GRV-4, which had Fisher parameters (Fisher 1953) for the VGP of $k < 6$. Also rejected were the anomalous mean poles from TXD-2 and KTM-2, which are displaced from the mean of the other Jurassic samples by more than 90°. For TXD-2 this is justified by the fact that TXD-6, from the same formation at a different locality, is in good agreement with the other Jurassic data (VGP'S), and the fact that TXD-2 samples were collected from a very small outcrop on the shore of Tuxedni Bay where slumping may have produced a local rotation. As explained above, no stable position was found for KTM-2, at the limit of demagnetization.

The four sites collected from Jurassic granitic batholiths (GRV-1, GRV-2, TXD-3, TXD-10) give mixed results (Fig. 10c). This may indicate that the assumption that present horizontal is identical to ancient horizontal is invalid. The only site with any positive attitude control is TXD-3, which is an outcrop on the south side of the Bruin Bay fault, overlain by gently dipping sediments. The gentle dip is thought to be due to the intrusive, thus the assumption is made that the batholith still has its original attitude. TXD-3 has been included in the overall means, but the other batholith sites, all from north of the fault, TXD-10, GRV-1, and GRV-2 have not.

Although there are other ways of combining the apparently stable results from individual sites by choosing different selection criteria, etc., they do not significantly alter the mean paleomagnetic pole position. The particular approach used here is the most viable paleomagnetically and geologically. Techniques such as a two-tier analysis (Watson and Irving 1957) were not applied to the data because the variances, k, are unequal, and because the products, K_iN_i, vary widely (McElhinny 1967; Bingham 1971).

As can be seen in Fig. 11, the distribution of the final means is elongated roughly along a great circle between the mean of means, present north, and the North American Jurassic VGP. There is no apparent correlation between position along this spread and the relative ages of the samples. The most plausible explanation is that not all of the secondary components aligned along the present field were removed; however, in most cases further 'cleaning' of the samples was not possible, usually because of rapidly decreasing intensities and increasing scatter.

A mean pole position and associated statistics have been calculated for the selected site means. It is recognized that the statistics are not strictly valid because of the noncircular distribution about the mean, nevertheless it is the most convenient way of displaying the data.

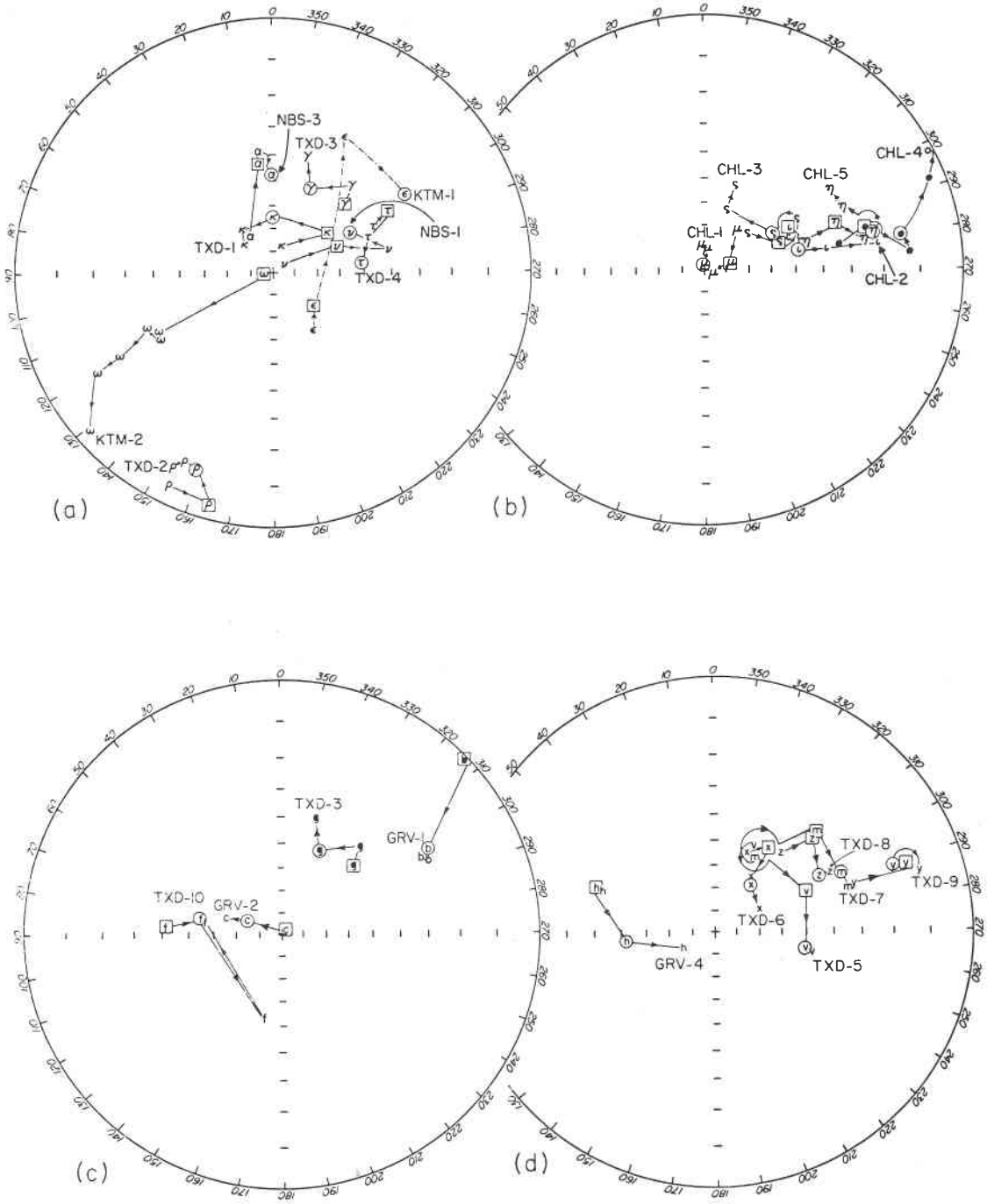


FIG. 10. Plots of mean VGP's for NRM P (NRM with respect to present horizontal) shown as ●→ , NRM (NRM with respect to ancient horizontal) shown as →□→, and demagnetization levels shown as →→ . The best demagnetization level for stable results is shown as →⊙→. Results from sediments are shown in (a), (b), and (d); results from batholiths are shown in (c).

TABLE 6. Jurassic paleomagnetic directions and pole positions for Alaska

SITE	TREATMENT	N	DIRECTION D	I	A95	K	POLE POSITION LAT	ELONG	A95	K
CHL-1	AF 95 OE	10	1	70	6.8	51.2	89	339	10.8	21.0
CHL-2	AF 95 OE	10	62	61	13.5	13.8	49	282	19.5	7.1
CHL-4	AF 95 OE	9	90	32	23.1	5.9	15	280	19.3	8.1
CHL-5	AF 95 OE	12	81	39	16.6	7.8	23	284	18.0	6.8
TXD-1	AF 95 OE	15	14	53	15.2	7.3	65	0	15.2	7.3
TXD-3	AF 190 OE	10	36	45	18.4	7.8	50	336	20.5	6.5
TXD-4	AF 95 OE	9	71	66	15.8	11.5	52	276	20.4	7.3
TXD-5	AF 95 OE	22	85	74	6.0	27.4	51	258	10.7	9.4
TXD-6	AF 190 OE	25	33	61	6.9	18.8	64	323	9.8	9.7
TXD-7	AF 95 OE	18	73	48	5.6	39.5	34	295	6.4	30.2
TXD-8	AF 95 OE	13	67	51	11.6	13.7	41	298	15.4	8.3
TXD-9	AF 95 OE	8	88	32	15.1	14.4	18	290	12.3	21.3
NBS-1	AF 95 OE	14	63	65	7.4	30.2	54	297	10.6	15.1
NBS-3	AF 95 OE	11	26	40	4.7	93.8	47	1	4.8	91.4
(INCL. BATHOLITH) MEAN		14					50	298	14.2	8.8
(SEDIMENTS ONLY) MEAN		13					50	295	14.9	8.8

NOTE: For explanation, see notes to Table 1.

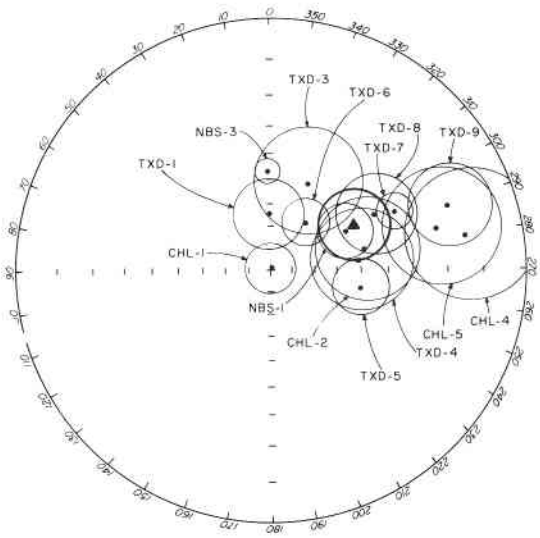


FIG. 11. Selected results and their mean shown with 95% circles of confidence.

Bearing in mind the problem with the precision parameters, it is found that the mean pole position of the 14 Jurassic VGP's remaining after selection, have a '*k*' equal to 8.8 and an α_{95} ofg 14.2° (Table 6). On the basis that the data are derived from sites that have a geographic spread of over 700 miles (1127 km), that have a varied lithology which includes sandstone, siltstone, and a granitic intrusive, and that the Jurassic period is adequately represented by rocks ranging in age from lower to upper Jurassic, the mean is considered a reasonably good representation of the Jurassic field for southern Alaska, with a probable bias towards the present pole.

Interpretation

Comparing the Alaskan data, which are all from the western limb of the Alaskan orocline, with those from elsewhere in North America, it is immediately seen that they are displaced from each other in the wrong direction to be explained by either Carey's (1958) or Grantz's (1966) Alaskan orocline hypotheses. This is true for both the overall mean and individual site means; thus the first conclusion drawn from the data is that the orocline concept for the tectonic development of southern Alaska is invalid.

For the purpose of comparing southern Alaskan data with North America, a compilation of Jurassic paleomagnetic poles was made, and is given in Table 7. Results from west of the Sierra Nevada, the Cascade Range, and the Coast Ranges appear to have been affected by the mobile belt formed by a long history of plate interaction. For this reason only results east of this belt have been included in the mean Jurassic North American pole. The mean paleomagnetic pole for the Jurassic from southern Alaska is shown in Fig. 12 along with selected

TABLE 7. Selected Jurassic paleomagnetic directions and pole positions for North America

GJ/NO	LOCATION	ROCK NAME	AGE	N	DIRECTION D	I	A95	K	POLE POSITION NLAT	ELONG	(DM,DP)	A95	REF
---	49.8N,63.2W	ANTICOSTI DIABASE DIKES	JL	11	10	57			76	85			Larochelle 1971
9/61	44N,71W	WHITE MTN VOLCANICS	JL	130	358	59	4	110	85	126	5,6		Opdyke & Wensink 1966
9/47	33.5-41.5N,72-83.5W	APPALACHIAN DIKES	J	223					66	145	3,3		de Boer 1967
3/26	39N,109W	CARMEL FORMATION	J		349	63	9	10	80	200	11,14		Collinson & Runcorn 1960
---	39.9N,110.1W	SUMMERVILLE FORMATION	JU	40	354	37	8	10	70	86	5,9		Steiner & Helsley 1972
3/30	37N,111.5W	KAYENTA FORMATION	JL	146	4	50		36	83	39			Collinson & Runcorn 1960
---	38.5N,109.6W	KAYENTA FORMATION*	JL	26	350	18			59	92		15	Steiner & Helsley 1972
	MEAN OF ABOVE SITES			7					77	107		11	

* UPPER & MIDDLE KAYENTA FORMATION OF STEINER & HELSLEY (1972) COMBINED

NOTES:
Column 1 gives the Geophysical Journal reference number of poles.
Column 2 gives the location of the sampling sites by geographical coordinates.
Column 3 gives name of the rock unit sampled.
Column 4 gives relative age of the rock unit as Jurassic (J) followed by U, M, or L, indicating Upper, Middle, or Lower, respectively.
Column 5 gives population used in statistical computations.
Columns 6 and 7 give the mean declination, in degrees east of true north, and mean inclination, in degrees below (+) or above (−) horizontal, after correction for known local tectonic movement.
Columns 8 and 9 give the radius in degrees of the α_{95} circle of confidence about the mean paleofield direction and the Fisher (1953) parameter, k.
Columns 10 and 11 give the coordinates of the paleomagnetic pole position in degrees north latitude and east longitude.
Columns 12 and 13 give the estimates of error, in degrees, of the pole position in either DM, DP, the semi-axes of the elliptical error at the 95% confidence level, or A95, the radius of the circular error at the 95% confidence level.
Column 14 gives the reference from which these data were taken.

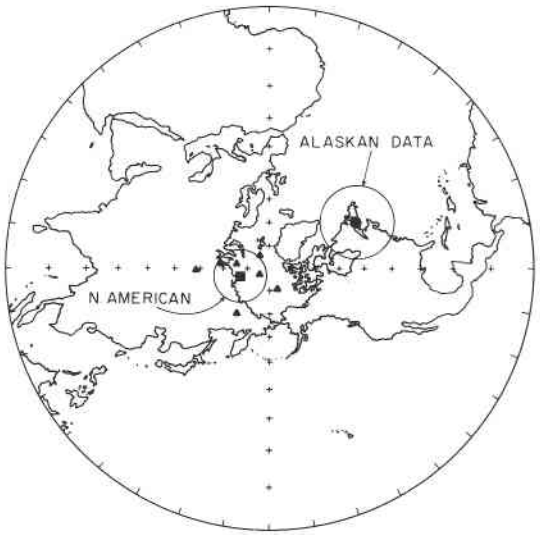


FIG. 12. Jurassic pole positions. Dot is mean of Alaskan data; square is mean of North American poles (triangles) from sites east of the trend of the Sierra Nevada. Circles are 95% confidence limits.

Jurassic paleomagnetic poles for North America and their mean. The Jurassic paleomagnetic pole from southern Alaska is significantly different from the present magnetic dipole and from the Jurassic paleomagnetic pole for North America.

Alaska's orientation and latitude in the Jurassic with respect to the present has been determined from the Jurassic paleomagnetic pole position by employing the common assumption that the average geomagnetic field is that of an axial dipole. For convenience, a reference site has been chosen, located at 60°N and 27°E, which is the location of Tuxedni Bay and also nearly the median of the Jurassic sampling areas. The Jurassic paleolatitude of the reference site has been determined to be 41°N, compared with the present latitude of 60°N, thus requiring a position approximately 19° further south. A counterclockwise rotation to 59° is also required to place Alaska in its Jurassic orientation. Applying the 95% confidence limits, Alaska could have been as far north as 55°N and as far south as 26°N. The 95% confidence limits require a minimum rotation of 40° with a maximum of 77°.

By superimposing the Jurassic paleomagnetic poles for Alaska and North America, their relative latitudes may be found. Doing this, it is found that southern Alaska had a paleolatitude equivalent to that of Oregon, and the continental margins off the Alaska Peninsula and off the Washington–Oregon coast were approximately parallel. If, as the parallelism perhaps suggests, Alaska and the west coast of America were adjacent in the Jurassic, then the required motion, determined graphically, can be described in terms of an opening of approximately 90° about an Euler pole at 60°N, 235°E. The paleolongitude of southern Alaska, however, is indeterminable from a single paleomagnetic pole position.

Polar wandering curves (Fig. 13) taken

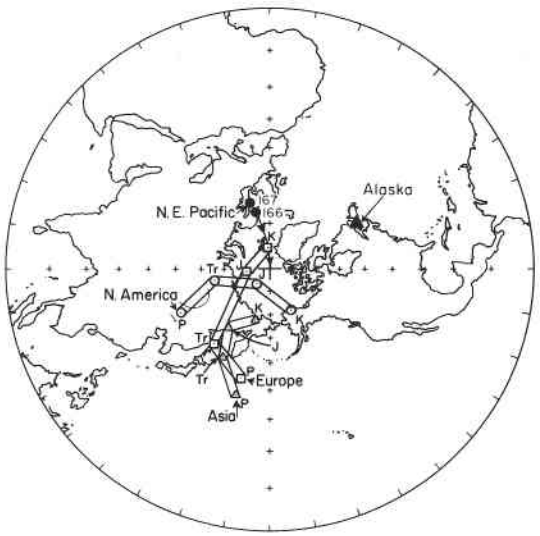


FIG. 13. Schematic representation of polar wandering curves for North America ○, Asia △, and Europe ⊡. Northeast Pacific Cretaceous pole ●, with trend →← . Mean Jurassic paleomagnetic pole ▲ for Alaska.

from Packer (1972) are not well enough defined at the 95% confidence level to show small movements of the poles, but show the trends of the motions involved. The schematic picture (Fig. 13) combining these polar wandering trends clearly shows the contrast between the Jurassic pole from southern Alaska and the poles for Permian through Cretaceous from North America, Asia, and Europe. A polar wander path for Alaska which includes the single Jurassic pole, must have the same trend as the results from the northeast Pacific.

Combining all this leads to a purely speculative model for the tectonic development of southern Alaska, which has southwestern Alaska in the Jurassic looking much like Baja California looks today. Between the Jurassic and the early Tertiary, Baja Alaska moved north and rotated clockwise to achieve its present position. This model is presently being tested by attempting to define the rest of the polar wander curve. The implications in terms of both macro- and micro-plate tectonics are the subject of a separate paper in preparation, with special reference to postulated movements of the Pacific with respect to North America. (See for instance, Atwater 1970; Larson and Chase 1972; and Clague and Jarrard 1973.)

Acknowldgements

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